

## Problem 5 — Full Walkthrough

### Symmetric rank-one matrix approximation

**Setup:** Let  $A = x_0 x_0^T \neq 0$  be a symmetric rank-one matrix,  $x_0 \in \mathbb{R}^n$ . Consider

$$f(x) := \frac{1}{2} \|A - xx^T\|_F^2, \quad x \in \mathbb{R}^n.$$

**A fact we'll use throughout:** the Frobenius norm is a norm, so  $\|A - xx^T\|_F^2 \geq 0$  always, with equality if and only if  $A - xx^T = 0$ , i.e.  $xx^T = A$  exactly. So  $f(x) \geq 0$  for every  $x$ , with  $f(x) = 0$  exactly when  $xx^T = A$ .

### Step 1 — Expand $f(x)$ into a usable formula

Since  $A$  and  $xx^T$  are both symmetric,  $(A - xx^T)^T = A - xx^T$ , so

$$\|A - xx^T\|_F^2 = \text{tr}((A - xx^T)^T(A - xx^T)) = \text{tr}((A - xx^T)(A - xx^T)).$$

Expand the product inside the trace:

$$(A - xx^T)(A - xx^T) = A^2 - Axx^T - xx^T A + xx^T xx^T.$$

Apply  $\text{tr}(\cdot)$  term by term, using that trace is invariant under cyclic permutation of a product ( $\text{tr}(PQ) = \text{tr}(QP)$ ):

$$\text{tr}(Axx^T) = \text{tr}(xx^T A) \quad \implies \quad \text{tr}(Axx^T) + \text{tr}(xx^T A) = 2\text{tr}(Axx^T).$$

Also, by the cyclic property,  $\text{tr}(Axx^T) = \text{tr}(x^T Ax) = x^T Ax$  (the trace of a  $1 \times 1$  matrix is just its entry). And

$$xx^T xx^T = x(x^T x)x^T = \|x\|^2 xx^T \quad \implies \quad \text{tr}(xx^T xx^T) = \|x\|^2 \text{tr}(xx^T) = \|x\|^2 \cdot \|x\|^2 = \|x\|^4,$$

using  $\text{tr}(xx^T) = x^T x = \|x\|^2$ . Putting it together:

$$\|A - xx^T\|_F^2 = \text{tr}(A^2) - 2x^T Ax + \|x\|^4 \quad \implies \quad f(x) = \frac{1}{2} \text{tr}(A^2) - x^T Ax + \frac{1}{2} \|x\|^4.$$

**Specialize to  $A = x_0 x_0^T$ :**  $\text{tr}(A^2) = \text{tr}(x_0 x_0^T x_0 x_0^T) = \|x_0\|^2 \text{tr}(x_0 x_0^T) = \|x_0\|^2 \cdot \|x_0\|^2 = \|x_0\|^4$ . And  $x^T Ax = x^T x_0 x_0^T x = (x_0^T x)(x_0^T x) = (x_0^T x)^2$  (since  $x^T x_0$  and  $x_0^T x$  are the same scalar). So:

$$f(x) = \frac{1}{2} \|x_0\|^4 - (x_0^T x)^2 + \frac{1}{2} \|x\|^4.$$

## Part (a) — Is the problem convex?

Test along  $x = s u_0$  where  $u_0 := x_0/\|x_0\|$  (unit vector) and  $s \in \mathbb{R}$ . Then  $x_0^T x = s x_0^T u_0 = s\|x_0\|$ , and  $\|x\|^2 = s^2$ , so:

$$f(su_0) = \frac{1}{2}\|x_0\|^4 - s^2\|x_0\|^2 + \frac{1}{2}s^4.$$

Differentiate twice with respect to  $s$ :

$$\frac{d}{ds}f(su_0) = -2s\|x_0\|^2 + 2s^3, \quad \frac{d^2}{ds^2}f(su_0) = -2\|x_0\|^2 + 6s^2.$$

At  $s = 0$ :  $\frac{d^2}{ds^2}f(su_0)|_{s=0} = -2\|x_0\|^2 < 0$  (since  $x_0 \neq 0$ ). A function with negative second derivative at a point is locally concave there — the opposite of what convexity requires. **So  $f$  is not convex, and the problem is not a convex optimization problem.**

## Part (b) — Find all stationary points

### Step 1 — Compute the gradient

Work with  $f(x) = \frac{1}{2}\text{tr}(A^2) - x^T A x + \frac{1}{2}\|x\|^4$  directly (the general form before specializing, since the gradient formula is cleaner this way).

**Gradient of  $x^T A x$ :** write  $x^T A x = \sum_i \sum_j x_i A_{ij} x_j$ . Differentiate with respect to one coordinate  $x_k$ :

$$\frac{\partial}{\partial x_k} \left( \sum_i \sum_j x_i A_{ij} x_j \right) = \sum_j A_{kj} x_j + \sum_i x_i A_{ik} = (Ax)_k + (A^T x)_k = 2(Ax)_k,$$

using  $A^T = A$  (given symmetric). So  $\nabla(x^T A x) = 2Ax$ .

**Gradient of  $\|x\|^4$ :**  $\|x\|^4 = (x^T x)^2$ . By the chain rule,  $\nabla[(x^T x)^2] = 2(x^T x) \cdot \nabla(x^T x) = 2\|x\|^2 \cdot 2x = 4\|x\|^2 x$ , so  $\nabla(\frac{1}{2}\|x\|^4) = 2\|x\|^2 x$ .

Combining:

$$\nabla f(x) = -2Ax + 2\|x\|^2 x = 2(\|x\|^2 I - A)x.$$

### Step 2 — Set the gradient to zero

$$\nabla f(x) = 0 \iff Ax = \|x\|^2 x.$$

So either  $x = 0$ , or  $x$  is an eigenvector of  $A$  with eigenvalue exactly  $\|x\|^2$ .

### Step 3 — The eigenstructure of $A = x_0 x_0^T$

For any  $v \in \mathbb{R}^n$ :  $Av = x_0(x_0^T v)$ . Two cases:

- If  $v \perp x_0$  (i.e.  $x_0^T v = 0$ ):  $Av = 0$ . So every vector orthogonal to  $x_0$  is an eigenvector with eigenvalue 0 — this eigenspace has dimension  $n - 1$ .
- If  $v = x_0$ :  $Ax_0 = x_0(x_0^T x_0) = \|x_0\|^2 x_0$ . So  $x_0$  is an eigenvector with eigenvalue  $\|x_0\|^2$ .

These two eigenspaces (dimension  $n - 1$  and dimension 1) together span all of  $\mathbb{R}^n$ , so this is the complete eigenstructure of  $A$ : eigenvalue 0 (multiplicity  $n - 1$ ) and eigenvalue  $\|x_0\|^2$  (multiplicity 1).

#### Step 4 — Solve $Ax = \|x\|^2x$ using this structure

**Case  $x = 0$ :** trivially satisfies the equation ( $0 = 0$ ). **Stationary point.**

**Case  $x \neq 0$  in the 0-eigenspace ( $x \perp x_0$ ):** then  $Ax = 0$ , so the equation becomes  $0 = \|x\|^2x$ . Since  $x \neq 0$ ,  $\|x\|^2 \neq 0$ , so this forces  $0 \neq 0$  — a contradiction. **No solutions here.**

**Case  $x = cx_0$  for some scalar  $c \neq 0$ :** then  $Ax = cAx_0 = c\|x_0\|^2x_0$ . We need this to equal  $\|x\|^2x = \|cx_0\|^2(cx_0) = c^3\|x_0\|^2x_0$ . Matching:  $c\|x_0\|^2x_0 = c^3\|x_0\|^2x_0 \Rightarrow c = c^3 \Rightarrow c(1 - c^2) = 0$ . Since  $c \neq 0$ , this forces  $c^2 = 1$ , i.e.  $c = \pm 1$ . **Stationary points:**  $x = x_0$  and  $x = -x_0$ .

**Case  $x = cx_0 + w$ , with  $c \neq 0$  and  $w \perp x_0$ ,  $w \neq 0$  (a genuinely mixed vector):** then  $x_0^T x = c\|x_0\|^2$  (since  $x_0^T w = 0$ ), so  $Ax = x_0(x_0^T x) = c\|x_0\|^2x_0$  — this has *no component along  $w$* . But the right-hand side of the stationary equation,  $\|x\|^2x = \|x\|^2(cx_0 + w)$ , *does* have a component along  $w$  (namely  $\|x\|^2w$ ), since  $w \neq 0$  and  $\|x\|^2 \neq 0$  (as  $x \neq 0$ ). Since  $x_0$  and  $w$  are linearly independent, the two sides can't match unless that  $w$ -component vanishes — but it doesn't. **No solutions here.**

#### Conclusion of Part (b)

Stationary points:  $x = 0$ ,  $x = x_0$ ,  $x = -x_0$ .

#### Part (c) — Is every local optimum a global optimum?

##### Step 1 — Evaluate $f$ at $x_0$ and $-x_0$

Recall from the preliminary remark:  $f(x) \geq 0$  always, with  $f(x) = 0$  exactly when  $xx^T = A$ . Check:  $x_0x_0^T = A$  exactly (that's literally how  $A$  was defined), so  $f(x_0) = 0$ . Similarly  $(-x_0)(-x_0)^T = x_0x_0^T = A$ , so  $f(-x_0) = 0$  too. Since  $f \geq 0$  everywhere and these points achieve the value 0 — the smallest value  $f$  can possibly take —  $x_0$  and  $-x_0$  are **global minima** (hence automatically local minima too, with no further calculus needed).

##### Step 2 — Evaluate $f$ at 0

$$f(0) = \frac{1}{2}\|x_0\|^4 - 0 + 0 = \frac{1}{2}\|x_0\|^4 > 0$$

(strictly positive since  $x_0 \neq 0$ ). This is *not* the global minimum value (0), so  $x = 0$  is not a global minimum. But is it perhaps still a local minimum?

##### Step 3 — Check whether $x = 0$ is even a local minimum

**Direction 1:** along  $u_0 = x_0/\|x_0\|$ . From Part (a):

$$f(su_0) - f(0) = \left[ \frac{1}{2}\|x_0\|^4 - s^2\|x_0\|^2 + \frac{1}{2}s^4 \right] - \frac{1}{2}\|x_0\|^4 = s^2 \left( \frac{1}{2}s^2 - \|x_0\|^2 \right).$$

For small  $|s|$  (specifically  $|s| < \sqrt{2}\|x_0\|$ ), the bracket  $(\frac{1}{2}s^2 - \|x_0\|^2)$  is negative, so  $f(su_0) - f(0) < 0$  for small  $s \neq 0$ .  $f$  **strictly decreases moving away from 0 in the  $u_0$  direction.**

**Direction 2:** along a unit vector  $w \perp x_0$ . Then  $x_0^T(sw) = s(x_0^T w) = 0$ , so:

$$f(sw) = \frac{1}{2}\|x_0\|^4 - 0 + \frac{1}{2}\|sw\|^4 = \frac{1}{2}\|x_0\|^4 + \frac{1}{2}s^4 = f(0) + \frac{1}{2}s^4 \geq f(0).$$

$f$  strictly increases moving away from 0 in any direction orthogonal to  $x_0$ .

So  $f$  decreases in one direction from 0 and increases in another — this is precisely the signature of a **saddle point**, not a local minimum (a local minimum would require  $f$  to not decrease in *any* direction).  $x = 0$  is **not a local minimum of  $f$** .

### Conclusion of Part (c)

The complete list of stationary points is  $\{0, x_0, -x_0\}$ . Of these:  $x_0$  and  $-x_0$  are global minima (Step 1), and 0 is not a local minimum at all (Step 3) — it's a saddle point, so it doesn't count as a “local optimum” that could fail to be global. Since every point that *is* a local minimum among the stationary points turns out to already be a global minimum, and there are no other candidates to check:

**Yes — every local optimum of this problem is a global optimum. ■**